MatchMind

Natural Language Processing-Powered Recruitment

Y Combinator Project Proposal

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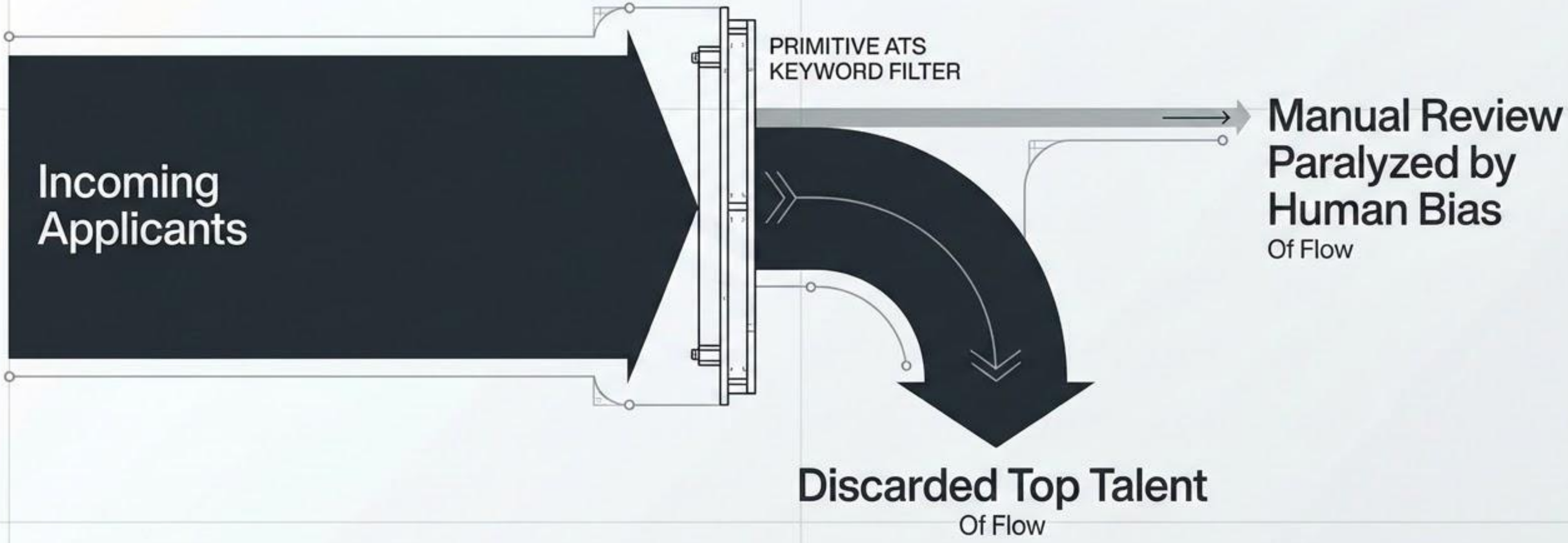
MatchMind replaces keyword-based filtering with Similarity based Ranking of talent in any applicant pool, eliminating manual screening bottlenecks.

The Cost of Vacancy

Lost HR Hours + Qualified Candidates Discarded = Massive Cost of Vacancy



The Intelligent Matching Engine

Vector-based Matching Over Keywords

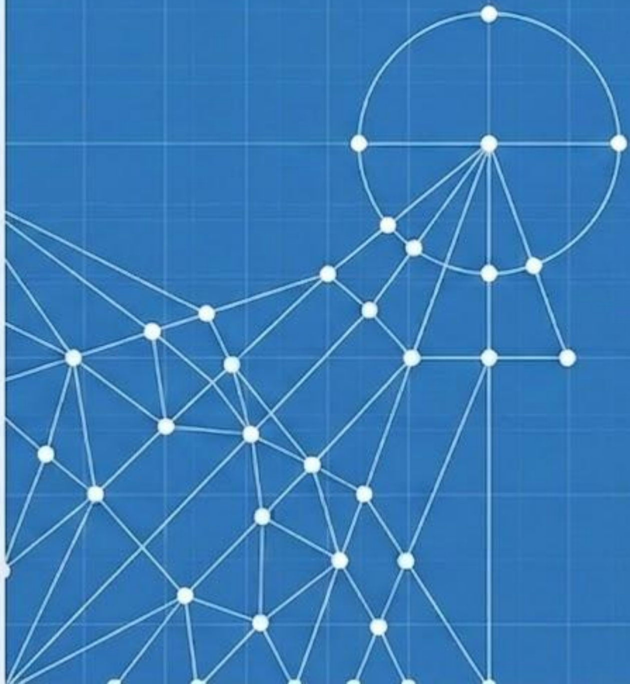
An intelligent matching engine that analyzes the cosine similarity of a Job Description, moving beyond simple keyword matching.

Objective Scale

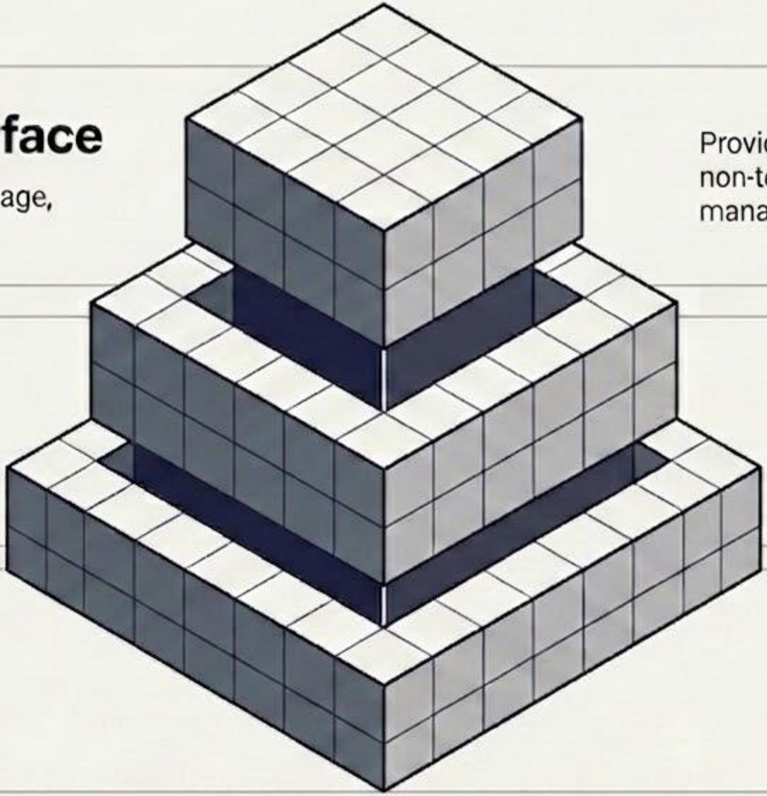
Provides instant, objective ranking of high-volume applicant pools based on actual skill alignment.

The Ultimate Outcome

Human Resources departments completely bypass the traditional resume pile and exclusively interview top-tier talent.



Technology Stack Analysis



Frontend User Interface

Bootstrap, Hypertext Markup Language, Cascading Style Sheets

Provides an intuitive layer ensuring non-technical Human Resources staff can manage high-volume data effortiessly.

Backend Server Architecture

Python, Flask

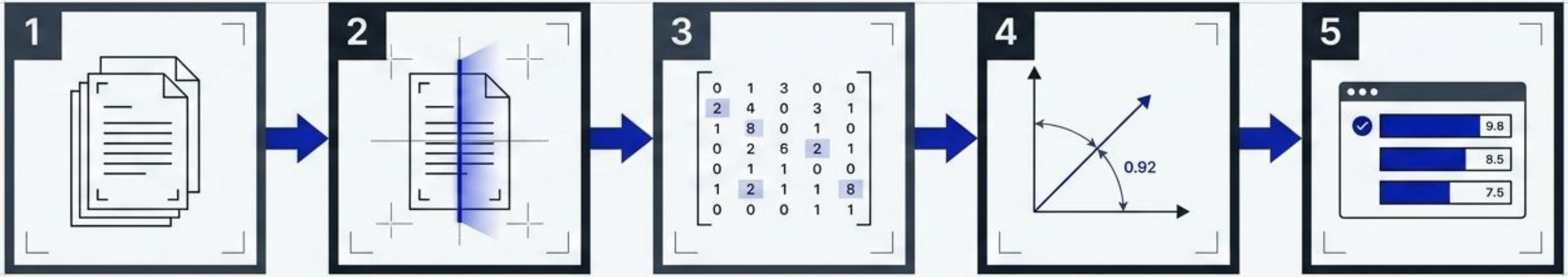
Enables rapid iteration of the ranking algorithm and lightweight, high-performance server management.

Data Parsing & Machine Learning

Scikit-learn, pdfplumber, docx2txt

Utilizes pdfplumber and docx2txt for high-fidelity text recovery. Scikit-learn, the industry standard for production-ready Term Frequency-Inverse Document Frequency implementation and mathematical scoring.

System Architecture & Workflow



Input Phase

High-volume batch CV uploads + Job Description

Text Extraction

Normalizing PDF/DOCX/TXT into a clean corpus

NLP Feature Extraction

TF-IDF vectorization isolates unique skill signatures

Mathematical Scoring

Cosine Similarity calculates multi-dimensional alignment

Result Presentation

Instant ranking rendered on a high-performance UI

Core Module Breakdown

Module 4: Ranking & UI
Zero-latency Flask/Bootstrap delivery to human operators.

ZERO-LATENCY DATA FLOW

Module 3: Similarity Calculation
High-speed semantic proximity engine executing multi-dimensional comparisons.

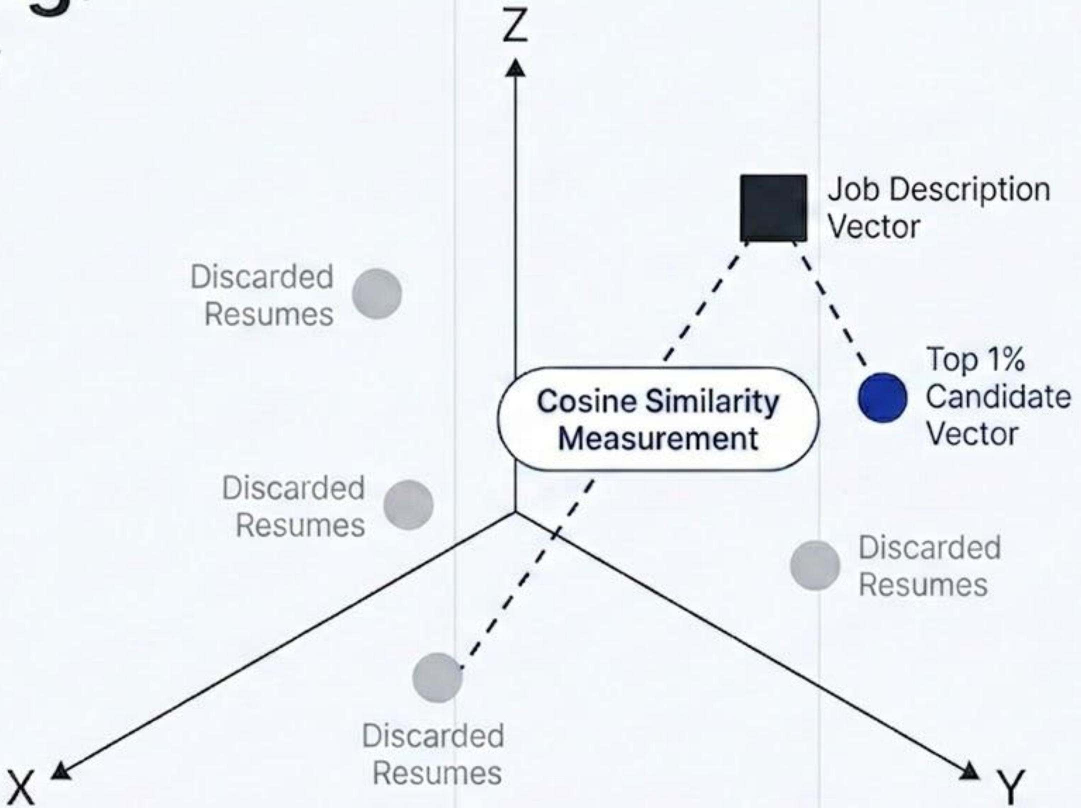
Module 2: Vectorization
Translates raw text into numerical representations of high-value soft and technical skills.

ZERO-LATENCY DATA FLOW

Module 1: Text Extraction
Utilizes pdfplumber and docx2txt for high-fidelity recovery of unstructured CV data.

The Math of Hiring: Vector Proximity

Moving beyond keyword matches to evaluate TF-IDF signatures via **Cosine Similarity**, measuring the exact conceptual distance between a job's requirements and a candidate's profile.

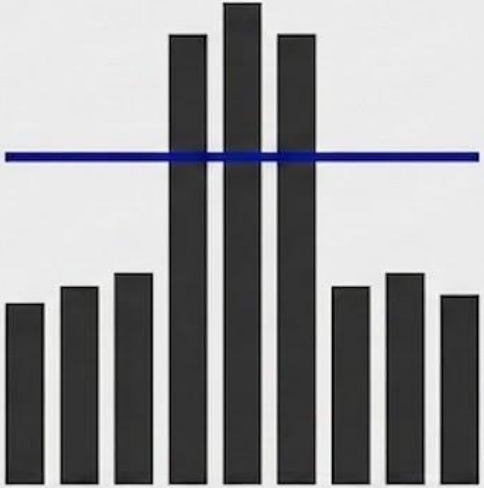


Technical Performance Targets

Human-System Alignment

Targeting expert-level precision matching.

Processing Throughput



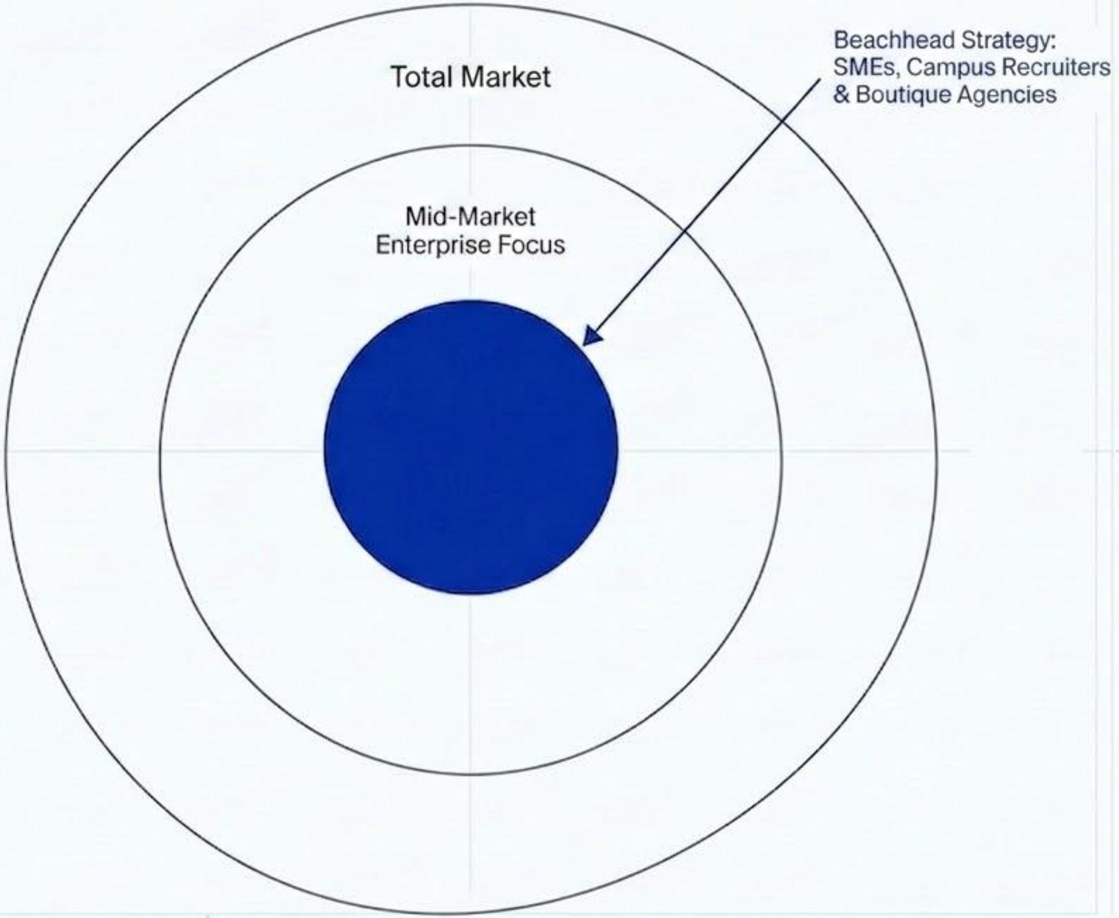
Maintains stability during peak campus hiring season spikes.

Extraction Fidelity

Flawless text recovery across multi-column PDF/DOCX layouts.

The AI-in-HR Opportunity

The post-pandemic landscape demands bias-free, enterprise-grade intelligence without enterprise costs. We are aggressively targeting the middle market who face the highest volume-to-resource pressure and are ignored by legacy platforms.



Market Beachhead and Revenue Generation

Target Market

Aggressively capturing the **Middle Market**—specifically **Boutique Staffing Agencies** and **Campus Recruiters** facing immense volume-to-resource pressure, which is our intended target Market.

Software as a Service Subscription

Tiered monthly deployment plans ensuring consistent platform usage for internal teams.

Enterprise Application Integration

Direct architectural integration into existing Enterprise Resource Planning platforms for massive scale.

Pay-per-Scan Credits

On-demand usage credits designed to capture high-volume seasonal spikes like campus hiring cycles.

The Competitive Edge

	Legacy Systems	Generic Chatbots	Manual Review	MatchMind
Vector-based Matching	✗			✓
Objective High-Volume Ranking		✗	✗	✓
Zero-Hallucination Mathematics		✗	✗	✓
Small and Medium Enterprise Accessibility	✗			✓

Versus Legacy Systems:
Replaces expensive, slow platforms that rely entirely on keyword density and miss the candidate's actual intent.

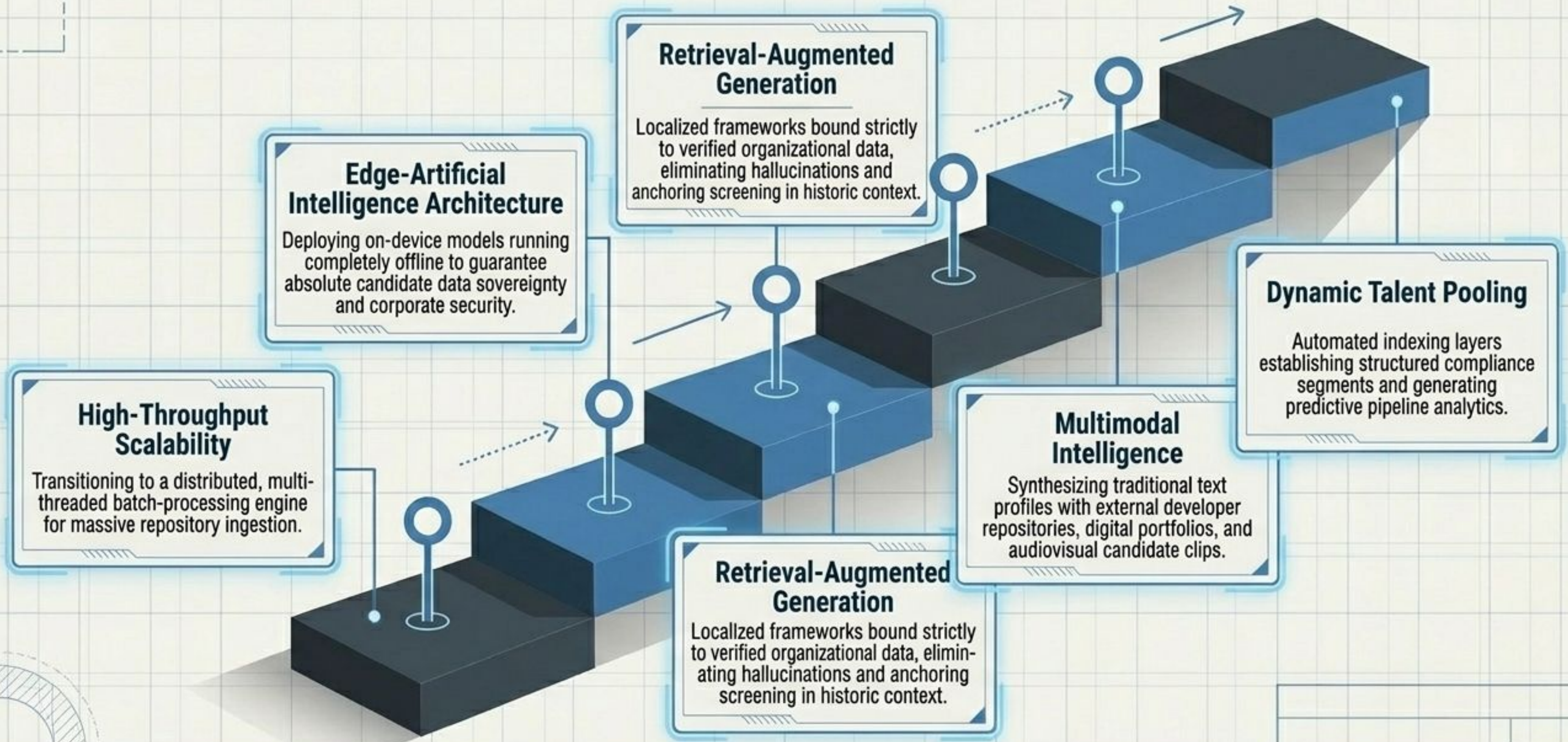
Versus Generic Chatbots:
Eliminates the severe hallucination risks of generic models through structured, purpose-built batch processing and strict mathematical scoring.

Versus Manual Review:
Transforms notoriously slow and biased human filtering into instant, data-driven ranking.

Recruitment Paradigm Matrix

Approach	Target Market Focus	Context Engine	Processing Risk	Deployment Speed
MatchMind	**SME-Friendly**	**Vector Similarity Ranking**	**Zero-Hallucination Math Scoring**	**Rapid**
Workday / Taleo	Prohibitively Expensive	Keyword-Heavy	Misses Semantic Context	Months/Years
Generic LLMs (ChatGPT)	General Purpose	Advanced	Hallucination Risks	Lacks Batch Structuring
Manual Review	Any	Subjective Human Parsing	High Bias & Inconsistency	Slowest

Roadmap and Future Scalability



Tech Stack Analysis

Backend Core

Python / Flask

Enables rapid iteration of the ranking algorithm and lightweight server management.

Frontend Interface

Bootstrap / HTML / CSS

Provides a clean, intuitive UI to ensure non-technical HR staff can manage high-volume data.

Machine Learning Libraries

Scikit-learn

Industry-standard for production-ready mathematical scoring and TF-IDF implementation.

Data Ingestion

pdfplumber / docx2txt / TXT

Reliable extraction of text from standard business document layouts.

Future State Architecture

BERT / S-BERT Embedding and Agentic Framework

Planned migration to deep-learning models (BERT and S-BERT) to capture nuances and sentence semantics, integrated within an Agentic Framework for adaptive, self-improving, and multi-step reasoning capabilities. This advances pure statistical vectors to full semantic intent and goal-oriented actions.

The Founding Team

Aggressive Computer Science and Data Science domain expertise rooted in applying NLP to real-world labor market inefficiencies. Bridging the gap between abstract ML research and modern HR workflows.

M Imran Mian
Head of Strategic and Compliance Initiatives

Afshan Ramzan
Security and Core Architect

Dania Saleem
Principal AI/ML Research Scientist

Hafsah Fatima
Chief Growth Officer (CGO)

MatchMind: Efficiency in every hire.